

based AR apps cannot work properly. At the same time, an error in any other sensor would result in a shift of the augmentation layer (depending of what has gone kaput).

## Recognition based AR

This is the most interesting breed on the list. The technologies that form the base here can aid other AR implementations. It also happens to be the most researched area for the same reason. Technologies involved in recognition based AR are as vivid as the use-cases. Recognition based AR depends on semantic understanding of the object in front of the camera and then replacing it with another object on the output display, or overlaying extra information related to the object.

The multidimensional usage of recognition systems involve various disciplines and quite a significant part of that (if not all) is useful for AR. Before we start off, it would be good to keep in mind that not all technologies pertaining to recognition systems apply to a single recognition based AR implementation.

### Image registration

Image registration is a process of merging different sets of data of the same object into one image. The different data sets can be different images of the same object taken at different times, from different angles and may include data received from other sources such as infrared view. When these different sets of data are merged into one, it may reveal a number of attributes which are not visible normally.

The process of registration involves more than one images. One of the images is called the target image and



The process of Image Registration allows the faint mountain range in the background to be recognised